Bhavya Mukherjee

EV Systems Engineer

RESUME_1823

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Total Experience: 15 years

PROFESSIONAL SUMMARY

EV Systems Engineer with 15 years of experience in ev systems, using Battery Management Systems, CAN Bus, MATLAB, Simulink. Experienced in model-based design, automotive safety and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: EV Powertrain, HIL, AUTOSAR, Simulink, Python, ISO 26262, Battery Management Systems, SOC Estimation, MATLAB, Motor Control, Diagnostics, CAN Bus

Methods: Model-based design, Automotive safety, Vehicle networks

PROFESSIONAL EXPERIENCE

Lead EV Systems Engineer | HelioNex Labs | Jan 2021 - Present

- Validated 228 HIL fault-injection scenarios for protective responses
- Lowered peak powertrain losses by 32% in control-model experiments
- Improved SOC estimation error by 26% after parameter calibration
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

EV Systems Engineer | NovaPeak Engineering | Jan 2016 - Dec 2020

- Improved SOC estimation error by 13% after parameter calibration
- Reduced simulated fault-detection delay by 41% using diagnostic timing refinements

Associate EV Systems Engineer | WillowGrid Labs | Jul 2011 - Dec 2015

- Validated 595 HIL fault-injection scenarios for protective responses
- Lowered peak powertrain losses by 34% in control-model experiments

PROJECTS

Charging Control Study | Python, MATLAB, Battery Management Systems, ISO 26262

- Simulated voltage and current control under load variations.
- Documented acceptance criteria and evaluated the design using automotive safety and vehicle networks.

Vehicle Network Diagnostics | ISO 26262, CAN Bus, Simulink, SOC Estimation

- Decoded CAN frames and generated fault trace reports.
- Documented acceptance criteria and evaluated the design using automotive safety and vehicle networks.

EDUCATION

B.Tech in Electrical Engineering, Crescent Valley Technical University, 2011

Relevant Methods: Model-based design, Automotive safety, Vehicle networks

CERTIFICATIONS

Battery Management Workshop (2019)